Majorization uncertainty bounds for incompatible binary measurements cannot be saturated

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Abstract

Majorization turns the uncertainty principle into a single state-independent probability vector that bounds every Schur-concave measure of uncertainty at once. We determine what this universality costs for two binary measurements. We prove that the optimal majorization vectors are attained by no state unless the measurements commute, so every bound they imply for a strictly Schur-concave measure, including the Shannon entropy, is strict; only the Rényi entropies of order zero and infinity are bounded tightly. For two qubit bases the direct-sum Shannon bound falls up to 22% short of the exact minimum, yet beats the Maassen–Uffink relation unless the bases are nearly mutually unbiased. We obtain the optimal bound for noisy and biased measurements in closed form and in any finite dimension, and show that detector bias, unlike detector noise, can invalidate ideal-detector uncertainty relations and the steering tests built on them; a single inequality decides when this happens.

Introduction

The uncertainty principle is one of the defining features of quantum mechanics[1–3]: some pairs of measurements cannot both have predictable outcomes, however carefully the system is prepared. This preparation uncertainty, distinct from the measurement uncertainty that limits joint measurements[4], has become a quantitative resource. It underlies the security of quantum key distribution[5–7], gives experimentally accessible criteria for entanglement[6,8,9] and for Einstein–Podolsky–Rosen steering[10–12], and is closely related to the incompatibility of measurements[13,14]. These applications need relations for measurements with finitely many outcomes whose bounds do not depend on the state. Robertson's variance relation[2] does not provide them; entropies of the outcome distributions do[15–19], the best known example being the Maassen–Uffink (MU)

relation[20], and entropic uncertainty relations are now a broad field with many applications and experimental tests[21–26].

Each entropic relation, however, constrains a single measure of uncertainty, whereas different tasks call for different measures: the Shannon entropy quantifies average information, the min-entropy governs guessing probabilities[27], and the Rényi and Tsallis families interpolate between these and other extremes. Majorization[28], the order that captures when one distribution is more disordered than another and a central tool of quantum information theory[29–31], treats all of them at once. One seeks a single state-independent vector that majorizes the outcome statistics of every state; every uncertainty measure that respects majorization then inherits a bound from this one vector. The programme was initiated by Partovi[32] and developed into universal uncertainty relations by Friedland, Gheorghiu and Gour[33] and by Puchała, Rudnicki and Życzkowski[34]; Rudnicki *et al.* introduced the stronger direct-sum form[35]. Majorization uncertainty relations have since been extended to generalized measurements, mixed states and quantum operations[36–38], made optimal in the majorization lattice[39–41], tested experimentally[41,42] and turned into steering witnesses[43].

What does this universality cost? Partovi already remarked that the optimal majorization vector is in general not the statistics of any state, so that a "minimum-uncertainty state" need not exist in this framework[32]; the point was recently reiterated in a review[44]. Li and Qiao observed that the partial sums of the optimal qubit bound are usually reached by different states, and that its Shannon entropy falls short of the optimal Shannon bound in examples[45]. What has been missing, even in the simplest setting, is a proof of when the optimal bounds are attained, an account of what non-attainment implies measure by measure, and an understanding of how the bounds behave for the imperfect measurements performed in experiments.

Here we give this account for two binary (two-outcome) measurements, a setting in which every step can be carried out in closed form. Unless the measurements commute, no state attains either optimal majorization vector, so every bound it implies for a strictly Schur-concave measure is strict; only the Rényi orders zero and infinity are bounded tightly. The cost is substantial: for two qubit bases the direct-sum Shannon bound falls short of the exact minimum by up to 22%. Even so, it outperforms MU unless the measurement axes are within about nine degrees of orthogonal, and its ratio to the exact minimum tends to one, though only logarithmically, as the measurements approach compatibility. For noisy and biased measurements we obtain the optimal direct-sum bound in closed form, for qubits and in any finite dimension. This bound has an immediate practical consequence. Detector noise can only increase uncertainty, so bounds derived for ideal detectors remain valid, but detector bias—produced, for instance, by mismatched detection efficiencies or by click/no-click detection—can push the statistics of a state outside the ideal quantum set. The single inequality $\Lambda \leq 2\sqrt{c}$, between a quantity of the measured effects and the overlap of the ideal ones, decides whether the ideal relation survives; when it fails, an uncertainty-based steering test can certify steering for a product state.

Several ingredients are known, and we credit them where they are used: the optimal bounds for two qubit bases are special cases of Refs. 33–35 and their optimality and saturating states were found by Li and Qiao[40]; the achievable expectation values fill an ellipse[46–48]; the direct-sum bound for unbiased unsharp qubit observables was evaluated by Baek *et al.*[37]; and the

relation between overlap and anticommutator in any dimension is due to Kaniewski, Tomamichel and Wehner[47]. New are the non-attainment theorems for both bounds and their consequences across the Rényi and Tsallis families, including an explicit counterexample to the direct-sum form at large orders; the closed-form optimal bound for biased measurements and its attainment conditions; the criterion for imperfect detectors and its consequence for steering tests; and the extension of these results to binary measurements in any finite dimension.

Results

Majorization uncertainty relations

For $x \in \mathbb{R}^n$ let $x^\downarrow$ denote its entries in non-increasing order and $L_k(x) = \sum_{i=1}^k x_i^\downarrow$ its Lorenz partial sums. For x, y with equal total, x is majorized by y, $x \prec y$, if $L_k(x) \le L_k(y)$ for all k[28]. A function Φ is Schur-concave if $x \prec y$ implies $\Phi(x) \ge \Phi(y)$, and strictly so if, in addition, $\Phi(x) > \Phi(y)$ whenever $x^\downarrow \neq y^\downarrow$. We use the Rényi and Tsallis entropies[49,50]

$$H_\alpha(p) = \frac{\log_2 \sum_i p_i^\alpha}{1-\alpha}, \qquad T_\gamma(p) = \frac{1 - \sum_i p_i^\gamma}{\gamma - 1}, \tag{1}$$

$\alpha \in [0, \infty]$, $\gamma > 0$, defined by continuity at $\alpha \in \{0, 1, \infty\}$ and $\gamma = 1$, with the Shannon entropy $H = H_1$ as common limit. H_∞ and $H_{1/2}$ are the min- and max-entropy[27]. All are Schur-concave; H_α is strictly Schur-concave for $0 < \alpha < \infty$ and T_γ for all $\gamma > 0$, whereas H_0 and H_∞ are not.

Let $p(\rho)$ and $q(\rho)$ be the outcome distributions of measurements M and N on a state ρ. A majorization uncertainty relation[32–35] is a state-independent vector ω with

$$p(\rho) \oplus q(\rho) \prec \omega \qquad \text{or} \qquad p(\rho) \otimes q(\rho) \prec \omega \qquad \forall \rho, \tag{2}$$

the direct-sum and tensor-product forms. Every Schur-concave Φ inherits $\Phi(p \oplus q) \ge \Phi(\omega)$ or $\Phi(p \otimes q) \ge \Phi(\omega)$. Sums of Shannon or Tsallis entropies are Schur-concave functions of $p \oplus q$; sums of Rényi entropies are Schur-concave functions of $p \otimes q$, since $H_\alpha(p \otimes q) = H_\alpha(p) + H_\alpha(q)$.

On non-negative, non-increasing vectors of fixed length and total, $\prec$ is a complete lattice[51,52]. We call ω the *optimal* majorization bound for a set $\mathcal{A}$ if it is the least upper bound $\sup \mathcal{A}$ in this lattice.

Lemma 1 (Refs. 51,52)**.** *Let $S_k = \sup_{a \in \mathcal{A}} L_k(a)$, $S_0 = 0$. The Lorenz curve of $\sup \mathcal{A}$ is the least concave majorant of $k \mapsto S_k$; in particular, if $\omega_k = S_k - S_{k-1}$ is non-increasing, then $\omega = \sup \mathcal{A}$.*

S_k is the largest value that any state can give to the summed probabilities of k outcomes. In every case below the increments are already ordered.

Binary qubit measurements and the achievable set

A qubit state is $\rho = \frac{1}{2}(\mathbb{1} + \vec{r} \cdot \vec{\sigma})$, $|\vec{r}| \le 1$. Two projective measurements along Bloch axes $\hat{m}$ and $\hat{n}$ give

$$p = \left(\tfrac{1+x}{2}, \tfrac{1-x}{2}\right), \qquad q = \left(\tfrac{1+y}{2}, \tfrac{1-y}{2}\right), \qquad x = \hat{m} \cdot \vec{r},\ y = \hat{n} \cdot \vec{r}, \tag{3}$$

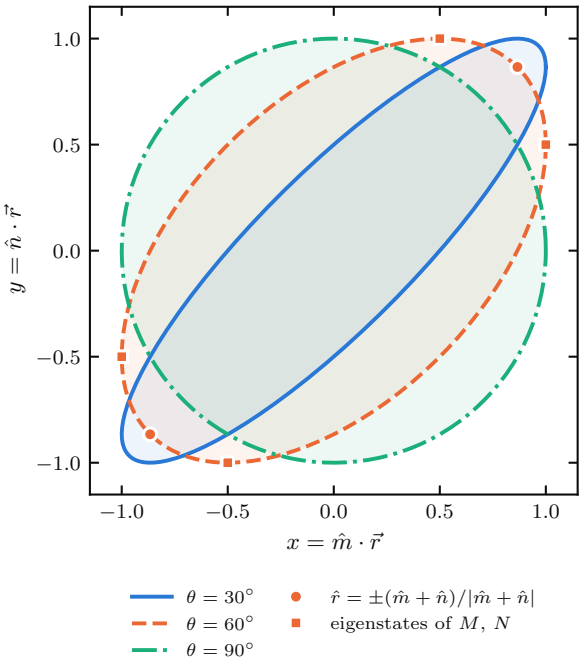


Figure 1: **Achievable expectation values for two qubit measurements.** The ellipse $\mathcal{E}_\theta$ of achievable $(x, y) = (\langle \hat{m} \cdot \vec{\sigma} \rangle, \langle \hat{n} \cdot \vec{\sigma} \rangle)$ for three Bloch angles θ. Markers ($\theta = 60°$) show the two families of states that saturate the partial sums of $\omega_\oplus$: eigenstates of M or N (squares; $k = 1, 3$) and bisector states $\hat{r} \propto \pm(\hat{m} + \hat{n})$ (circles; $k = 2$). No single point saturates all partial sums (Proposition 1).

and their maximal overlap is

$$c = \max_{i,j} |\langle m_i | n_j \rangle|^2 = \tfrac{1}{2}\big(1 + |\hat{m} \cdot \hat{n}|\big) = \cos^2 \tfrac{\theta}{2}, \tag{4}$$

where $\theta \in [0, \pi/2]$ is the acute Bloch angle between the axes; relabelling outcomes, we take $\hat{m} \cdot \hat{n} = \cos\theta \geq 0$. A general two-outcome POVM $\{E_0, \mathbb{1} - E_0\}$ is

$$E_0 = \tfrac{1}{2}\big[(1 + \mu)\mathbb{1} + \eta\, \hat{a} \cdot \vec{\sigma}\big], \qquad \eta \geq 0, \quad |\mu| + \eta \leq 1, \tag{5}$$

with sharpness η and bias μ; projective measurements have $\eta = 1$, $\mu = 0$.

Lemma 2 (Refs. 46–48)**.** *For non-parallel axes,* $\{(\hat{m} \cdot \vec{r}, \hat{n} \cdot \vec{r}) : |\vec{r}| \leq 1\}$ *is the filled ellipse*

$$\mathcal{E}_\theta = \{(x, y) : x^2 - 2xy\cos\theta + y^2 \leq \sin^2\theta\}, \tag{6}$$

whose boundary consists of the pure states with Bloch vectors in the plane of the two axes, and $\max_{|\vec{r}| \leq 1}(sx + ty) = |s\hat{m} + t\hat{n}|$ *for all real* s, t.

The achievable statistics $\{p(\rho) \oplus q(\rho)\}$ are thus an affine image of $\mathcal{E}_\theta$ (Fig. 1). Moving a point radially outwards makes both p and q more ordered, so every measure $\Phi(p) + \Psi(q)$ with Φ, Ψ Schur-concave is minimized on $\partial\mathcal{E}_\theta$. Equivalently, $(x, y) \in \mathcal{E}_\theta$ iff the covariance matrix of $A = \hat{m} \cdot \vec{\sigma}$ and $B = \hat{n} \cdot \vec{\sigma}$ is positive semidefinite; its off-diagonal entry contains the state-independent anticommutator $\frac{1}{2}\{A, B\} = \cos\theta\, \mathbb{1}$ (Methods).

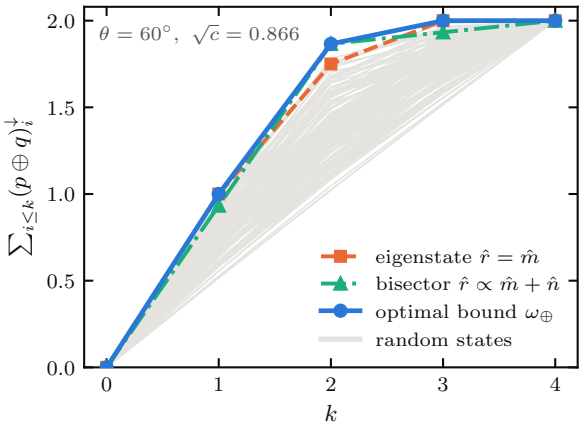


Figure 2: **The optimal bound is an envelope, not a state.** Lorenz curves of $p \oplus q$ for $\theta = 60°$. Grey: 150 random states. The optimal bound $\omega_\oplus$ (blue) is the smallest concave curve above all of them; it touches the eigenstate curve at $k = 1, 3$ and the bisector-state curve at $k = 2$, but no single state reaches it.

Optimal bounds and their non-attainment

Theorem 1 (Refs. 35,40)**.** *For projective qubit measurements with maximal overlap c,*

$$p(\rho) \oplus q(\rho) \prec \omega_\oplus := \left(1,\ \sqrt{c},\ 1-\sqrt{c},\ 0\right) \qquad \forall \rho, \tag{7}$$

and $\omega_\oplus$ is the optimal majorization bound.

Theorem 2 (Refs. 32–34)**.** *Under the same assumptions, $p(\rho) \otimes q(\rho) \prec \omega_\otimes := (w, 1-w, 0, 0)$ for all ρ, with $w = \frac{1}{4}(1+\sqrt{c})^2$, and $\omega_\otimes$ is the optimal majorization bound.*

Both follow from Lemma 2 in a few lines (Methods). The central partial sum of $\omega_\oplus$ is $S_2 = 1 + \sqrt{c}$, the qubit case of the Landau–Pollak-type inequality $p_i + q_j \le 1 + \sqrt{c}$ behind Deutsch's relation[18,53]; for mutually unbiased bases $\omega_\otimes$ is the vector of Partovi[32]. The two families of states that saturate the partial sums never coincide:

Proposition 1. *(i) A state with $(p \oplus q)^{\downarrow} = \omega_\oplus$ exists iff $c = 1$, i.e., iff M and N commute. For $c < 1$, S_1 and S_3 are attained only by eigenstates of M or N, and S_2 only by the bisector states $\vec{r} = \pm(\hat{m} + \hat{n})/|\hat{m} + \hat{n}|$ (for $\theta = \pi/2$ also by $\pm(\hat{m} - \hat{n})/|\hat{m} - \hat{n}|$). (ii) A state with $(p \otimes q)^{\downarrow} = \omega_\otimes$ exists iff $c = 1$.*

Demanding certainty about M lowers the largest attainable $L_2(p \oplus q)$ from $1 + \sqrt{c}$ to $1 + c$, a deficit $\sqrt{c}(1 - \sqrt{c})$ that vanishes only for commuting measurements (Fig. 2). The optimal vectors are therefore suprema in the majorization lattice, not the statistics of any state, which makes precise, for the binary case, the general remark of Partovi[32]. Once the saturating states of the individual partial sums are known[40], the proof is short; its value lies in what follows from it.

Consequences for entropic uncertainty

Let $h(t) = -t\log_2 t - (1-t)\log_2(1-t)$. Theorems 1 and 2 give the known bounds[33–35]

$$H(p) + H(q) \geq h(\sqrt{c}), \qquad T_\gamma(p) + T_\gamma(q) \geq \frac{1 - c^{\gamma/2} - (1-\sqrt{c})^\gamma}{\gamma - 1}, \tag{8}$$

$$H_\alpha(p) + H_\alpha(q) \geq H_\alpha(\sqrt{c}, 1-\sqrt{c}) \quad (0 < \alpha \leq 1), \qquad H_\alpha(p) + H_\alpha(q) \geq H_\alpha(\omega_\otimes) \quad (\alpha \in [0, \infty]). \tag{9}$$

For $\alpha \leq 1$ the direct-sum bound is the stronger one; that the direct-sum form cannot be extended to all orders was stated in Ref. 35, and we make it explicit:

Proposition 2. *(i) For $0 < \alpha \leq 1$, $H_\alpha(\sqrt{c}, 1-\sqrt{c}) \geq H_\alpha(\omega_\otimes)$, with equality iff $c = 1$. (ii) For $c < 1$ the direct-sum form fails at $\alpha = \infty$: $\min_\rho[H_\infty(p) + H_\infty(q)] = -2\log_2 \frac{1+\sqrt{c}}{2} < -\frac{1}{2}\log_2 c$.*

The failure, however, sets in only beyond $\alpha = 2$. Numerically, for qubits $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ is a valid bound for every $\alpha \leq 2$ at every angle, and fails above an angle-dependent order between 2.00 ($\theta \to 0$) and 2.58 ($\theta \to 90°$). At $\alpha = 2$, where the bisector state is optimal at every angle[54], validity reduces to $(\sqrt{c}-1)^3(\sqrt{c}+3) \leq 0$ (Methods). The direct-sum form thus extends well beyond the range $\alpha \leq 1$ in which it is proven in general.

Non-attainment now has a sharp consequence.

Theorem 3. *Let $c < 1$. (i) For every continuous, strictly Schur-concave Φ, $\min_\rho \Phi(p \oplus q) > \Phi(\omega_\oplus)$ and $\min_\rho \Phi(p \otimes q) > \Phi(\omega_\otimes)$. Hence the Shannon and Tsallis bounds in Eq. (8) and the Rényi bounds in Eq. (9) with $0 < \alpha < \infty$ are all strict. (ii) The tensor bounds of order $\alpha = 0$ and $\alpha = \infty$ are tight, attained by eigenstates and by bisector states, respectively.*

The proof uses only that the achievable set is compact and that its supremum is not attained, so it applies verbatim to every case below where the optimal bound is not attained. As majorization bounds, $\omega_\oplus$ and $\omega_\otimes$ cannot be improved; as bounds on any single strictly Schur-concave measure, they always can. For such measures the optimum must be found measure by measure, which is what entropy-specific relations do.

How large is the gap? The exact minimum of $H(p) + H(q)$ for two qubit bases is known[55,56]: it lies on the great circle through $\hat{m}$ and $\hat{n}$, equals $2h(\frac{1+\sqrt{c}}{2})$ at the bisector state for $\theta \leq \bar{\theta}$, where $\sqrt{c}\,\mathrm{artanh}\,\sqrt{c} = 1$[57], i.e. $\bar{\theta} \approx 67.07°$, and moves continuously to the eigenstates, reached at $\theta = 90°$, for larger angles. Figure 3 compares it with the majorization bounds and with MU, $H(p) + H(q) \geq -\log_2 c$. Over all angles $h(\sqrt{c})$ is at least 78.2% of the exact minimum (worst at $\theta \approx 70°$; equivalently, the minimum exceeds the bound by up to 28%), the largest absolute gap being 0.196 bit at $\theta \approx 74°$; the tensor bound $h(w)$ reaches at least 76.1% of the minimum (at $\theta \approx 70.6°$) and falls short by up to 0.217 bit. At $\theta = 90°$, where MU is tight, $h(\sqrt{c}) = 0.872$ bit. Yet $h(\sqrt{c}) > -\log_2 c$ for all $\theta < \theta^* \approx 81.3°$ (79.3° for the tensor bound), the root of $h(\cos\frac{\theta}{2}) = -2\log_2\cos\frac{\theta}{2}$, in agreement with the numerical observation of Ref. 37. As $\theta \to 0$, with $\varepsilon = 1 - \sqrt{c} \approx \theta^2/8$, one finds $h(\sqrt{c}) \approx \varepsilon\log_2(e/\varepsilon)$ and an exact minimum $2h(\varepsilon/2) \approx \varepsilon\log_2(2e/\varepsilon)$, so their ratio tends to one as $1 + 1/\log_2(e/\varepsilon)$ (1.09 at $\theta = 0.1$ rad, 1.047 at 0.003 rad), whereas the ratio of $h(\sqrt{c})$ to MU grows as $\frac{1}{2}\ln(8e/\theta^2)$. The eigenstates, although they saturate S_1 and S_3, give $h(c)$, asymptotically twice the minimum. For the Shannon entropy

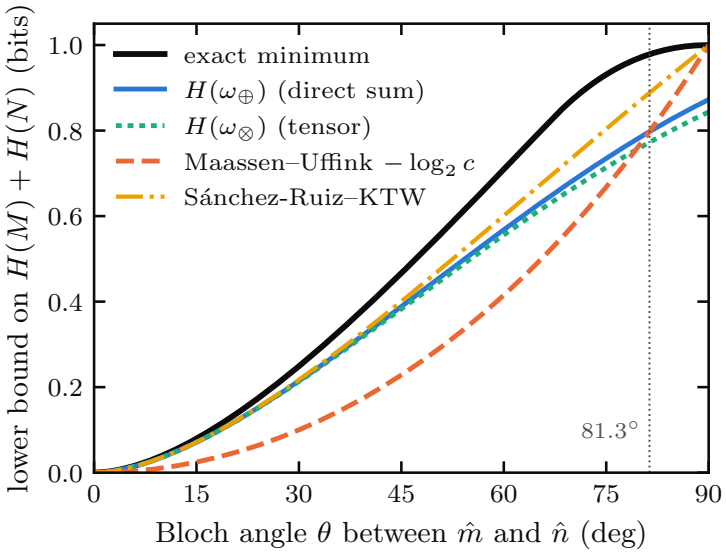


Figure 3: **The price of universality for the Shannon entropy.** Lower bounds on $H(M)+H(N)$ versus the Bloch angle θ ($c=\cos^2\frac{\theta}{2}$). Black: exact minimum, equal to $2h(\frac{1+\sqrt{c}}{2})$ for $\theta \le \bar{\theta} \approx 67.07°$. Blue: direct-sum bound $h(\sqrt{c})$. Green (dotted): tensor-product bound $h(w)$. Orange (dashed): Maassen–Uffink. Yellow (dash-dotted): the closed-form Shannon bound $h(\frac{1+\sqrt{2c-1}}{2})$[47,55], strictly above $h(\sqrt{c})$ and at least as strong as MU. The direct-sum bound beats MU for $\theta < 81.3°$ (dotted line).

alone, the closed-form bound[47,55] $h\big(\frac{1+\sqrt{2c-1}}{2}\big)$ is strictly stronger than $h(\sqrt{c})$ at every angle and at least as strong as MU, with which it coincides at 90° (0.601 against 0.568 bit for $h(\sqrt{c})$ at 60°); what majorization offers instead is one bound for all Schur-concave measures.

Figure 4 extends the comparison to the Rényi family. For $\alpha \le 1$ MU also applies, since H_α is non-increasing in α. At $\theta = 60°$ both majorization bounds beat MU for all $\alpha \le 1$; at $\theta = 85°$ only for $\alpha < 0.69$ (direct sum) and $\alpha < 0.58$ (tensor). For the max-entropy the direct-sum bound reaches 83% and 91% of the exact minimum at the two angles. For large α only the tensor bound remains, and it converges to the exact minimum as $\alpha \to \infty$, as Theorem 3 requires. Which state is optimal depends on the measure[54,58,59]: for $c > \frac{1}{2}$ the eigenstates are optimal for $\alpha \le \alpha_0(c)$ and the bisector for $\alpha \ge \alpha^\star(c)$, with intermediate states in between. Here α_0 exceeds $\frac{1}{2}$ (0.52, 0.56 and 0.66 at 20°, 60° and 85°), and $\alpha^\star$ rises from $\frac{1}{2}$ as $c \to 1$ to $\alpha^\dagger \approx 1.43$ as $c \to \frac{1}{2}$[54]; the bisector is optimal up to 43° for $\alpha = \frac{3}{4}$, up to $\bar{\theta}$ for the Shannon entropy, up to 80° for $\alpha = 1.2$ and at all angles for $\alpha \ge \alpha^\dagger$, consistent with the tightness of the min-entropy bound.

Noisy and biased measurements

For POVMs, the partial sums of the direct-sum relation are largest eigenvalues of sums of effects[35–37,60], $S_k = \max_{|J|=k} \lambda_{\max}\big(\sum_{j\in J} K_j\big)$, with the K_j running over the effects of both measurements; the flattened vector they define is optimal for general measurements[41]. For two-outcome qubit POVMs, whose achievable expectation values were described geometrically in Ref. 48, the maximization can be done in closed form for arbitrary bias, and no flattening is needed. Let M have parameters $(\mu, \eta, \hat{a})$ and N parameters $(\nu, \zeta, \hat{b})$ in Eq. (5).

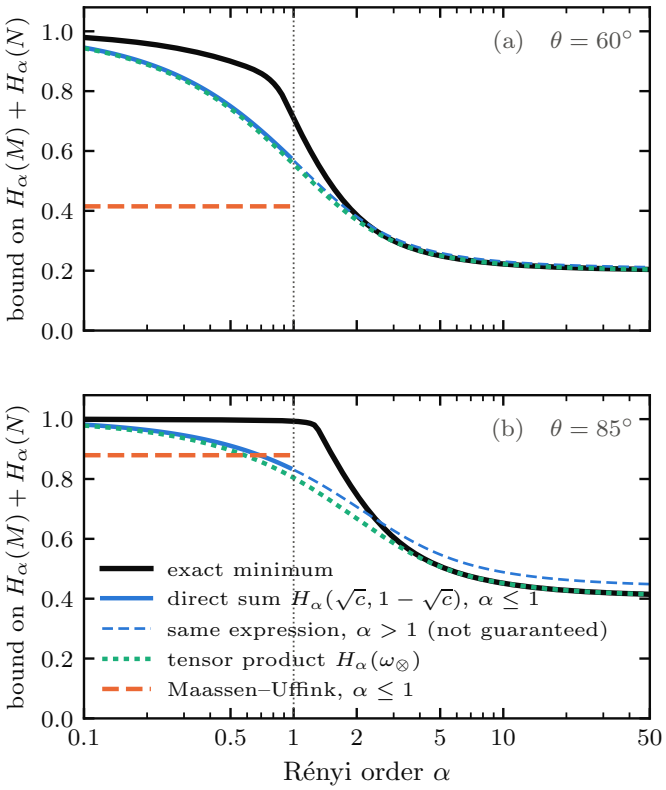


Figure 4: **Bounds across the Rényi family.** Bounds on $H_\alpha(M) + H_\alpha(N)$ for (a) $\theta = 60°$ and (b) $\theta = 85°$. Black: exact minimum. Blue: direct-sum bound $H_\alpha(\sqrt{c}, 1-\sqrt{c})$, proven for $\alpha \le 1$; its thin dashed continuation for $\alpha > 1$ remains a valid bound up to $\alpha = 2$ but exceeds the exact minimum above $\alpha \approx 2.1$ in (a) and $\alpha \approx 2.5$ in (b). Green (dotted): tensor-product bound $H_\alpha(\omega_\otimes)$. Orange (dashed): MU, valid for $\alpha \le 1$. Dotted vertical line: Shannon case $\alpha = 1$.

Theorem 4. *Define*

$$\kappa = \max\{|\mu| + \eta,\ |\nu| + \zeta\}, \qquad \Lambda = \max_{s,t=\pm 1} \left[s\mu + t\nu + |s\eta\hat{a} + t\zeta\hat{b}| \right]. \tag{10}$$

Then $p(\rho) \oplus q(\rho) \prec \omega_\oplus(M,N)$ for all ρ, where

$$\omega_\oplus(M,N) = \tfrac{1}{2}\big(1+\kappa,\ 1+\Lambda-\kappa,\ 1+\kappa-\Lambda,\ 1-\kappa\big) \tag{11}$$

is the optimal majorization bound; $S_1 = \frac{1}{2}(1+\kappa)$ and $S_2 = 1 + \frac{1}{2}\Lambda$.

For projective measurements $\kappa = 1$, $\Lambda = 2\sqrt{c}$, and Eq. (7) is recovered; for unbiased ones $\Lambda = (\eta^2 + \zeta^2 + 2\eta\zeta|\hat{a}\cdot\hat{b}|)^{1/2}$, in agreement with Ref. 37. With unequal sharpness or bias the relevant geometric quantities are the lengths $|\eta\hat{a} \pm \zeta\hat{b}|$ rather than c, and the bound is the qubit case of Corollary 2 below. Theorem 4 also covers states of fixed spectrum: projective measurements on states with $|\vec{r}| = r$ have the statistics of unbiased measurements of sharpness $\eta = \zeta = r$ on pure states, so the spectrum-dependent bound[38,40] is $\frac{1}{2}(1+r,\ 1-r+2r\sqrt{c},\ 1+r-2r\sqrt{c},\ 1-r)$, and by Proposition 3 below it is not attained for $r > 0$ and $c < 1$.

Proposition 3. *(i) If E_0 and F_0 do not commute, $\eta\zeta\,\hat{a}\times\hat{b} \neq 0$, no state attains $\omega_\oplus(M,N)$. (ii) If $\mu = \nu = 0$ and E_0, F_0 commute, a state attains it. (iii) With bias, commuting effects*

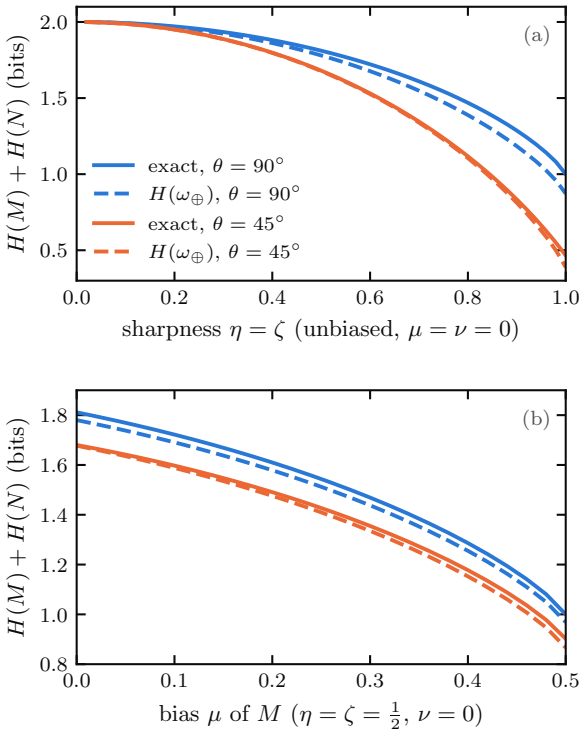


Figure 5: **Noisy and biased qubit measurements.** Exact minimum of $H(M) + H(N)$ (solid) and the bound from Theorem 4 (dashed) for Bloch angles 90° and 45°. (a) Unbiased measurements of equal sharpness ($\mu = \nu = 0$, $\eta = \zeta$). (b) M with sharpness $\eta = \frac{1}{2}$ and bias μ; N unbiased with $\zeta = \frac{1}{2}$.

need not attain it: for $\hat{b} = \hat{a}$, $(\mu, \eta) = (0.3, 0.6)$ *and* $(\nu, \zeta) = (-0.8, 0.15)$, S_1 *is attained only at* $\vec{r} = -\hat{a}$ *and* S_2 *only at* $\vec{r} = \hat{a}$.

By Theorem 3, $H(p) + H(q) \geq -\sum_k \omega_k \log_2 \omega_k$ with $\omega = \omega_\oplus(M, N)$ is therefore strict whenever the effects do not commute. Figure 5 shows that it tracks the exact minimum as functions of sharpness and of bias. The example in (iii), two readouts of the same axis favouring opposite ends of it, is a trade-off of purely classical origin. Attainment is also unrelated to joint measurability: unbiased POVMs are jointly measurable iff $|\eta\hat{a} + \zeta\hat{b}| + |\eta\hat{a} - \zeta\hat{b}| \leq 2$[13,61] (the biased criterion is more involved[62–64]); for orthogonal axes and $\eta = \zeta \leq 1/\sqrt{2}$ the pair is jointly measurable, yet $S_2 = 1 + \eta/\sqrt{2} < 1 + \eta$ and the bound is not attained. Majorization uncertainty relations thus quantify preparation uncertainty, which persists without measurement incompatibility.

Imperfect detectors and uncertainty-based tests

Bias and unequal sharpness are what real detectors produce. If the two outputs of a polarizing beam splitter are monitored by detectors of efficiencies η_H and η_V and no-click events are assigned to either outcome at random, the effect of outcome H has the form (5) with $\mu = (\eta_H - \eta_V)/2$ and $\eta = (\eta_H + \eta_V)/2$; a mismatch of efficiencies, a known concern in quantum key distribution[65], appears as a bias. A single click/no-click detector behind a polarizer, with no-click counted as the second outcome, gives $E_0 = \eta_d P$, i.e. $\mu = \eta_d - 1$, $\eta = \eta_d$ and $|\mu| + \eta = 1$. Experiments that test uncertainty relations, or use them to certify entanglement or steering,

usually compare data with the bound for ideal measurements. Theorem 4 tells exactly when this is justified.

Proposition 4. *Let M, N be two-outcome qubit POVMs with axes $\hat{a}, \hat{b}$, let M°, N° be the projective measurements along the same axes, with statistics p°, q° and overlap $c = \frac{1}{2}(1+|\hat{a}\cdot\hat{b}|) < 1$. (i) If $\mu = \nu = 0$, then $p(\rho)\oplus q(\rho) \prec p^\circ(\rho)\oplus q^\circ(\rho)$ for every ρ; hence every lower bound on $\Phi(p\oplus q)$ or on $\Phi(p)+\Psi(q)$, with Φ, Ψ Schur-concave, that holds for M°, N° also holds for M, N. (ii) In general, the ideal relation $p(\rho) \oplus q(\rho) \prec \omega_\oplus = (1, \sqrt{c}, 1-\sqrt{c}, 0)$ holds for all ρ iff $\Lambda \le 2\sqrt{c}$. If $\Lambda > 2\sqrt{c}$, some state violates the Landau–Pollak relation $p_i + q_j \le 1 + \sqrt{c}$ of the ideal measurements. (iii) If M and N are read out by the same detectors, $\mu = \nu$ and $\eta = \zeta$, with $\hat{a}\cdot\hat{b} \ge 0$, the condition reads $|\mu| \le \sqrt{c}\,(1-\eta)$. Click/no-click detection violates it for every $\eta_d < 1$.*

Noise only blurs the statistics towards the uniform distribution, but bias shifts the image of $\mathcal{E}_\theta$, and the shifted ellipse can leave the ideal one. Whether a particular entropic bound is violated depends on the measure: the Shannon minimum is more forgiving than the Landau–Pollak sum, but not immune (Fig. 6). For orthogonal axes and random no-click assignment, an efficiency $\eta_H = 0.99$ violates the ideal relation for $\eta_V < 0.942$ and the ideal Shannon bound of one bit for $\eta_V < 0.578$; for $\eta_H = 0.95$ the thresholds are 0.708 and 0.211, and for $\eta_H \le 0.9$ the Shannon bound is never violated, although the ideal relation still is for $\eta_V < 0.417$. Click/no-click detectors lower the Shannon minimum below its ideal value at all efficiencies and angles we examined; for orthogonal axes this holds for every $\eta_d < 1$, since the eigenstate of the dark outcome of M gives $H(M) + H(N) = h(\eta_d/2) < 1$. In all cases $H(\omega_\oplus(M,N))$ remains a valid bound.

This matters for steering tests[10–12], in which Bob trusts his measurements and Alice's are uncharacterized.

Corollary 1. *If Alice's measurements A_1, A_2 and Bob's two-outcome measurements M, N admit a local-hidden-state model, then*

$$H(M|A_1) + H(N|A_2) \;\ge\; \min_\rho \big[H(M) + H(N)\big] \;\ge\; H\big(\omega_\oplus(M,N)\big). \tag{12}$$

The minimum for the ideal measurements M°, N° may replace the right-hand side if M, N are unbiased, but not in general: for click/no-click detection along orthogonal axes, a product state with Bob in the dark eigenstate of M gives $H(M|A_1) + H(N|A_2) = h(\eta_d/2)$, below the ideal value of one bit for every $\eta_d < 1$.

An uncertainty-based steering or entanglement test with biased trusted detectors must therefore use the bound of the measurements actually performed; Theorem 4 provides it in closed form, for every Schur-concave measure at once, and Proposition 4 tells when the ideal bound can be kept.

Binary measurements in any finite dimension

Theorem 5. *Let $M = \{P, \mathbb{1} - P\}$ and $N = \{Q, \mathbb{1} - Q\}$ be projective measurements on $\mathbb{C}^d$, write $P_1 = P$, $P_2 = \mathbb{1} - P$, $Q_1 = Q$, $Q_2 = \mathbb{1} - Q$, and let $c = \max_{i,j} \|P_i Q_j\|^2$. Then Theorems 1 and 2 hold verbatim, and $\omega_\oplus$ and $\omega_\otimes$ are attained iff $c = 1$.*

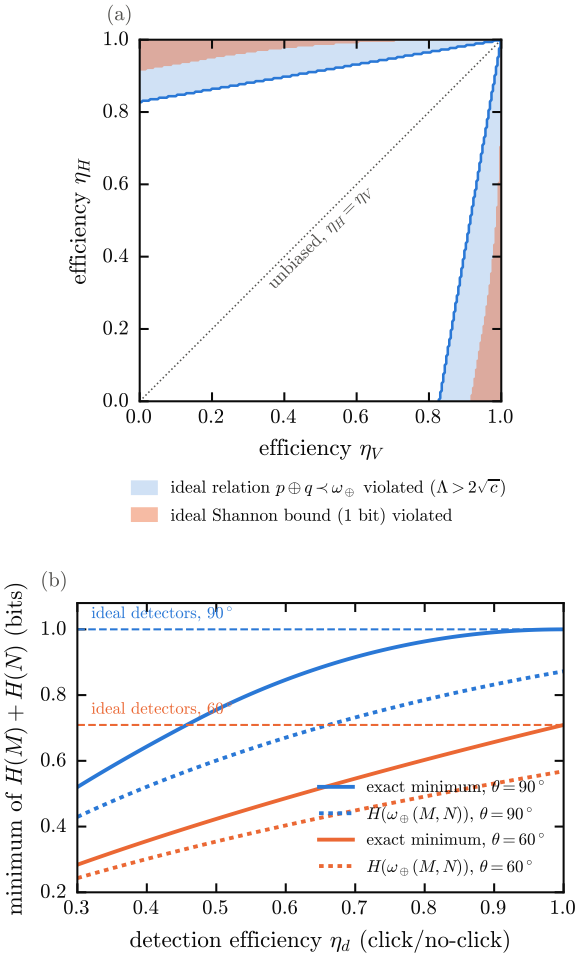


Figure 6: **Detector bias can invalidate ideal-detector uncertainty bounds.** (a) Polarizing beam splitter with detectors of efficiencies (η_H, η_V) and random assignment of no-click events, for measurements along orthogonal axes read out by the same detectors. Blue: the ideal relation $p \oplus q \prec \omega_\oplus$ fails ($\Lambda > 2\sqrt{c}$, Proposition 4); orange: the ideal Shannon bound of one bit fails. Unbiased detectors ($\eta_H = \eta_V$, dotted) never violate ideal bounds. (b) Click/no-click detection with efficiency η_d: exact minimum of $H(M) + H(N)$ (solid) and the bound $H(\omega_\oplus(M,N))$ of Theorem 4 (dotted) for Bloch angles 90° and 60°; dashed lines mark the minimum for ideal detectors.

The direct-sum part follows from the eigenvalue formula[37] and $\lambda_{\max}(P+Q) = 1 + \|PQ\|$[66], and the tensor part from the reduction to qubit blocks by Jordan's lemma[47,66]; new are the tensor bound and the attainment conditions in dimension d. The overlap has the operator form $c = \frac{1}{2} + \frac{1}{4}\|\{A, B\}\|$ with $A = 2P - \mathbb{1}$, $B = 2Q - \mathbb{1}$, as shown in any dimension by Kaniewski *et al.*[47], closely related to the formula $\|PQ + QP\| = \|PQ\| + \|PQ\|^2$[67]. All results of the previous sections therefore hold for binary projective measurements in any dimension; the min-entropy bound remains tight, and for $c < 1$ the bounds for strictly Schur-concave measures are strict. The direct-sum argument uses only the eigenvalue formula and carries over to noisy and biased measurements:

Corollary 2. *Let $M = \{E_1, E_2\}$ and $N = \{F_1, F_2\}$ be two-outcome POVMs on $\mathbb{C}^d$, let S_1 be*

the largest eigenvalue of the four effects and $S_2 = \max_{i,j} \lambda_{\max}(E_i + F_j)$. *Then*

$$p(\rho) \oplus q(\rho) \prec \big(S_1,\ S_2 - S_1,\ 1 + S_1 - S_2,\ 1 - S_1\big) \qquad \forall \rho, \tag{13}$$

and this vector is the optimal majorization bound.

Theorem 4 is the qubit case, $S_1 = \frac{1}{2}(1+\kappa)$ and $S_2 = 1 + \frac{1}{2}\Lambda$, and Theorem 5 the projective case, $S_1 = 1$ and $S_2 = 1+\sqrt{c}$. Part (ii) of Proposition 4 generalizes in the same way: since $S_1 \leq 1$ and $S_3 = 1 + S_1 \leq 2$ always, two binary POVMs obey the relation $p \oplus q \prec (1, \sqrt{c}, 1 - \sqrt{c}, 0)$ of a reference pair of projective measurements with overlap c iff $S_2 \leq 1 + \sqrt{c}$.

Discussion

For binary measurements the majorization approach to uncertainty can be carried out completely. The achievable statistics of a qubit are an affine image of an ellipse, and a single four-component vector, determined by the overlap c, is their least upper bound in the majorization lattice, for the direct sum as well as for the tensor product; for POVMs it is determined by κ and Λ, and in any dimension by the largest eigenvalues of the effects and of their pairwise sums. One vector controls every Schur-concave uncertainty measure at once.

The price of this unification is now precise. Because the supremum is not the statistics of any state unless the measurements commute, the induced bound is strict for every strictly Schur-concave measure and exact only at Rényi orders zero and infinity. Quantitatively the price is moderate where it matters most: the direct-sum Shannon bound is never below 78% of the exact value, beats MU for Bloch angles below 81.3°, and becomes asymptotically tight in ratio as the measurements approach compatibility, precisely where MU is weakest. Where only one measure matters, as for the Shannon entropy in the analysis of key distribution, the entropy-specific optimum should be used; where many matter at once, or where the measure is not fixed in advance, majorization gives the only universal bound, and we now know how far from tight it is.

The second lesson concerns imperfect measurements. Majorization makes the distinction between noise and bias sharp: noise acts as a doubly stochastic map on the statistics and can only increase every measure of uncertainty, whereas bias moves the achievable set. Whether ideal-detector relations survive is decided by one partial sum, S_2, that is, by $\Lambda \leq 2\sqrt{c}$ for qubits. Uncertainty-based tests of entanglement and steering with trusted detectors must therefore account for detector bias explicitly, as they do for losses in other contexts; the closed-form bound of Theorem 4 does so for all Schur-concave measures simultaneously. Non-attainment is itself testable: experiments have reported Lorenz curves that touch the optimal bound [42], which by Proposition 1 can only happen for individual partial sums, and the distance $\min_\rho \max_k [S_k - L_k(p \oplus q)]$ between the bound and the closest state is a directly measurable quantity. A natural next step is an experimental test with deliberately mismatched detectors, for instance in the coherent-light [42] or photonic [41] settings in which optimal majorization bounds have already been probed.

Relation to quantum optics. For the quadratures of a light mode, coherent and squeezed states saturate both the Heisenberg relation and the entropic relation of Białynicki-Birula and

Mycielski[17], and pure Gaussian states are conjectured to majorize every Wigner-positive state[68]. For non-commuting binary measurements no such state exists (Propositions 1 and 3), and the optimal state depends on the measure[58,59]. The boundary of $\mathcal{E}_\theta$ plays the role of the squeezed-state family, and the balanced bisector state that of a coherent state, which minimizes the Shannon sum up to $\bar{\theta} \approx 67°$. The difference is algebraic: for quadratures the commutator is a c-number, whereas for binary observables the anticommutator is, blockwise, a c-number, and it fixes $c = \frac{1}{2} + \frac{1}{4}\|\{A, B\}\|$. The Robertson bound vanishes for every state in the plane of the two axes, which contains all optimal states; the missing information is carried by the anticommutator term of the Robertson–Schrödinger relation, which every pure state saturates (Methods). For single photons, two linear polarization analysers at relative angle ϑ give $\theta = 2\vartheta$ and $c = \cos^2\vartheta$ (Malus's law); the crossover with MU and the angle $\bar{\theta}$ correspond to $\vartheta \approx 40.7°$ and $33.5°$.

Open questions. (i) The optimal tensor-product bound for noisy or biased measurements: already for unbiased measurements of equal sharpness S_3 is attained neither by eigenstates nor by bisector states. (ii) A closed form for $\lambda_{\max}(E_i + F_j)$ in Corollary 2, analogous to $1 + \|P_i Q_j\|$ for projectors, and attainment conditions in dimension d. (iii) More than two measurements: for qubit bases $S_k = \frac{k}{2} + \frac{1}{2}\max_J |\sum_{j\in J} \vec{n}_j|$, and Li and Qiao obtained the optimal bound for $\sigma_x, \sigma_y, \sigma_z$[40]; for random triples we always found the increments ordered, but have no proof. (iv) Whether state-independent relations of the form (2) can reproduce mixed-order relations such as $H_\infty(p) + H_{1/2}(q) \geq -\log_2 c$, which conditional majorization yields for qubits[69]. (v) Binary measurements on light modes: for photon-number parity and displaced parity, or for coherent-state tests realized by displacement and on/off detection, $c = 1$ without a common eigenvector, so no non-trivial state-independent relation exists, although individual partial sums remain informative; for coherent-state tests $\pi[Q(\nu_1) + Q(\nu_2)] \leq 1 + e^{-|\nu_1 - \nu_2|^2/2}$, with equality for an optical cat state[70,71], which plays the role of the bisector state. Non-trivial relations require $\|\{A, B\}\| < 2$, as for pseudospin operators[72] or polarization measurements on single photons.

Methods

Proof of Lemma 2 and the covariance form

Write $\vec{r} = \vec{r}_\parallel + \vec{r}_\perp$ with $\vec{r}_\parallel \in \mathrm{span}\{\hat{m}, \hat{n}\}$. Only $\vec{r}_\parallel$ contributes to $(x, y) = G\xi$, where ξ are the coefficients of $\vec{r}_\parallel$ in the basis $\{\hat{m}, \hat{n}\}$ and $G = \left(\begin{smallmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{smallmatrix}\right)$ is the Gram matrix. Then $|\vec{r}_\parallel|^2 = \xi^T G \xi = (x, y) G^{-1} (x, y)^T \leq 1$, which is Eq. (6) since $\det G = \sin^2\theta$; equality holds iff $\vec{r}_\perp = 0$ and $|\vec{r}| = 1$. Conversely every $(x, y) \in \mathcal{E}_\theta$ is attained by $\vec{r} = \xi_1 \hat{m} + \xi_2 \hat{n}$ with $\xi = G^{-1}(x, y)^T$. The support function is Cauchy–Schwarz: $sx + ty = \vec{r} \cdot (s\hat{m} + t\hat{n})$. The covariance matrix of A, B is $\Gamma = \left(\begin{smallmatrix} 1-x^2 & \cos\theta - xy \\ \cos\theta - xy & 1-y^2 \end{smallmatrix}\right)$ and $\det\Gamma = \sin^2\theta - (x^2 - 2xy\cos\theta + y^2)$, so $(x, y) \in \mathcal{E}_\theta$ iff $\Gamma \geq 0$. The Robertson–Schrödinger relation reads $\det\Gamma \geq \langle(\hat{m} \times \hat{n}) \cdot \vec{\sigma}\rangle^2$; for a pure state $\langle(\hat{m} \times \hat{n}) \cdot \vec{\sigma}\rangle^2 = \sin^2\theta - (x^2 - 2xy\cos\theta + y^2)$, so pure states saturate it, and in the plane of the axes it reduces to the boundary $\det\Gamma = 0$.

Proofs of Theorems 1 and 2

Direct sum. Clearly $S_1 \leq 1$ and $S_4 = 2$; since all entries are non-negative, $S_3 = 2 - \min_\rho \min_i (p \oplus q)_i \leq 2$, and both bounds are attained by an eigenstate of M. For $k = 2$ the two largest entries

either belong to the same distribution, with sum 1, or are of the form $p_i + q_j = 1 + \frac{1}{2}(\pm x \pm y) \leq 1 + \frac{1}{2}\max\{|\hat{m}+\hat{n}|, |\hat{m}-\hat{n}|\}$ by Lemma 2. With $\hat{m}\cdot\hat{n} \geq 0$ the maximum is $|\hat{m}+\hat{n}| = 2\cos\frac{\theta}{2} = 2\sqrt{c}$, so $S_2 = 1+\sqrt{c}$, attained by the bisector state. The increments of $(1, 1+\sqrt{c}, 2, 2)$ give $\omega_\oplus$, which is non-increasing because $1 \geq \sqrt{c} \geq 1 - \sqrt{c}$ (as $c \geq \frac{1}{2}$); Lemma 1 completes the proof.

Tensor product. By the arithmetic–geometric mean inequality and $S_2 = 1 + \sqrt{c}$, $p_i q_j \leq (\frac{p_i+q_j}{2})^2 \leq (\frac{1+\sqrt{c}}{2})^2 = w$, with equality at the bisector state, where $p_+ = q_+ = (1+\sqrt{c})/2$. Hence $S_1 = w$. For an eigenstate of M, $p \otimes q = (q_+, q_-, 0, 0)$, so $S_2 = S_3 = S_4 = 1$. Since $\sqrt{c} \geq 1/\sqrt{2}$, $w \geq \frac{1}{2}$ and the increments $(w, 1-w, 0, 0)$ are non-increasing.

Proof of Proposition 1

(i) $(p \oplus q)^{\downarrow}_1 = 1$ forces an eigenstate of M or N, say $\vec{r} = \pm\hat{m}$. Then $q = (\frac{1\pm\cos\theta}{2}, \frac{1\mp\cos\theta}{2})$ and $(p \oplus q)^{\downarrow} = (1, c, 1-c, 0)$, which equals $\omega_\oplus$ iff $c = \sqrt{c}$, i.e. $c = 1$. The statement about S_2 follows from the equality condition in Cauchy–Schwarz and from $|\hat{m} - \hat{n}| < |\hat{m} + \hat{n}|$ for $\theta < \pi/2$. (ii) If $c = 1$, a common eigenstate gives $p \otimes q = (1, 0, 0, 0) = \omega_\otimes$. If $c < 1$, $L_2(p \otimes q) = 1 = L_2(\omega_\otimes)$ requires two vanishing entries of $p \otimes q$, hence an eigenstate of M or N, at which $L_1(p \otimes q) = c < w = L_1(\omega_\otimes)$, since $w - c = \frac{1}{4}(1-\sqrt{c})(1+3\sqrt{c}) > 0$.

Proofs of the entropic bounds and Proposition 2

Eq. (8) is Karamata's inequality applied to Theorem 1 with $\phi(t) = -t\log_2 t$ and $\phi_\gamma(t) = (t - t^\gamma)/(\gamma-1)$, using $\sum_i p_i = \sum_j q_j = 1$. For the direct-sum Rényi bound with $\alpha < 1$, write $\sum_i p_i^\alpha = 1 + X$, $\sum_j q_j^\alpha = 1 + Y$ with $X, Y \geq 0$; then $H_\alpha(p) + H_\alpha(q) = \frac{\log_2[(1+X)(1+Y)]}{1-\alpha} \geq \frac{\log_2(1+X+Y)}{1-\alpha}$ and $1 + X + Y = \sum_k (p \oplus q)_k^\alpha - 1 \geq \sum_k (\omega_\oplus)_k^\alpha - 1 = c^{\alpha/2} + (1-\sqrt{c})^\alpha$ by Schur-concavity[35]. The tensor bounds follow from $H_\alpha(p) + H_\alpha(q) = H_\alpha(p \otimes q)$[33,34]; for $\alpha = \infty$ the bound $-2\log_2\frac{1+\sqrt{c}}{2}$ is attained at the bisector state and contains Deutsch's Shannon bound[18]. Proposition 2(i): since $w - \sqrt{c} = \frac{1}{4}(1-\sqrt{c})^2 \geq 0$, $(\sqrt{c}, 1-\sqrt{c}) \prec (w, 1-w)$ with equality iff $c = 1$; apply strict Schur-concavity. (ii): with $t = \sqrt{c}$, $(\frac{1+t}{2})^2 - t = (\frac{1-t}{2})^2 > 0$ for $t < 1$, and the minimum is the tight tensor bound. *Validity at* $\alpha = 2$. With the bisector optimal[54] and $s = \sqrt{c}$, $a = \frac{1+s}{2}$, the exact minimum is $-2\log_2[a^2 + (1-a)^2] = -2\log_2\frac{1+s^2}{2}$, and $H_2(s, 1-s) = -\log_2[s^2 + (1-s)^2]$. The bound holds iff $(\frac{1+s^2}{2})^2 \leq s^2 + (1-s)^2$, i.e. iff $\frac{1}{4}(s-1)^3(s+3) \leq 0$, which is true for $s \leq 1$.

Proof of Theorem 3

(i) The achievable statistics form a compact set, so the minimum of Φ is attained at some ρ^*. By Theorem 1 and Proposition 1(i), $p \oplus q(\rho^*) \prec \omega_\oplus$ with $(p \oplus q)^{\downarrow} \neq \omega_\oplus$, hence $\Phi(p \oplus q(\rho^*)) > \Phi(\omega_\oplus)$; the tensor case is analogous. The Shannon and Tsallis sums are strictly Schur-concave functions of $p \oplus q$, and $H_\alpha(p) + H_\alpha(q) = H_\alpha(p \otimes q)$ with H_α strictly Schur-concave for $0 < \alpha < \infty$. For the direct-sum Rényi bound with $\alpha < 1$, apply the argument to $v \mapsto \sum_k v_k^\alpha$. (ii) An eigenstate of M gives $H_0(p) + H_0(q) = 1 = H_0(\omega_\otimes)$, and a bisector state gives $\max_i p_i \max_j q_j = w$.

Proof of Theorem 4

The largest single outcome probability is $\max_\rho \frac{1}{2}(1 + |\mu + \eta\hat{a}\cdot\vec{r}|) = \frac{1}{2}(1 + |\mu| + \eta)$ for M, so $S_1 = \frac{1}{2}(1+\kappa)$; likewise the smallest is $\frac{1}{2}(1-\kappa)$, so $S_3 = 2 - \frac{1}{2}(1-\kappa)$. For S_2, a pair within

one distribution sums to 1, while a cross pair gives, by Cauchy–Schwarz, $\max_\rho\big[1+\frac{1}{2}\big(s(\mu+\eta\hat{a}\cdot\vec{r})+t(\nu+\zeta\hat{b}\cdot\vec{r})\big)\big]=1+\frac{1}{2}\big[s\mu+t\nu+|s\eta\hat{a}+t\zeta\hat{b}|\big]$. Choosing $s=\operatorname{sgn}\mu$, $t=\operatorname{sgn}\nu$ shows $\Lambda\geq 0$, so $S_2=1+\frac{1}{2}\Lambda$. The increments of (S_k) are the entries of Eq. (11); they are non-increasing iff $\kappa\leq\Lambda\leq 2\kappa$. The upper inequality follows from the triangle inequality, $\Lambda\leq|\mu|+|\nu|+\eta+\zeta\leq 2\kappa$. For the lower one, let ρ^* attain S_1 via an outcome of, say, M; then $S_2\geq S_1+\max_j q_j(\rho^*)\geq S_1+\frac{1}{2}$, i.e. $\Lambda\geq\kappa$. Lemma 1 completes the proof.

Proof of Proposition 3

(i) Suppose a state attains S_1 through an outcome of M, so that $\kappa=|\mu|+\eta$ (the case of N is symmetric). Since $\eta>0$, this forces $\vec{r}=s_0\hat{a}$ with $s_0\mu=|\mu|$. There the two largest entries of $p\oplus q$ sum to $1+\frac{1}{2}\max_t[|\mu|+\eta+t\nu+ts_0\zeta\,\hat{a}\cdot\hat{b}]$, whereas the term $s=s_0$ of Λ gives $S_2\geq 1+\frac{1}{2}[|\mu|+t\nu+|\eta\hat{a}+ts_0\zeta\hat{b}|]$ for each t. By Cauchy–Schwarz, $|\eta\hat{a}+ts_0\zeta\hat{b}|>\eta+ts_0\zeta\,\hat{a}\cdot\hat{b}$ unless $\hat{b}=\pm\hat{a}$ (and $\eta\pm ts_0\zeta\geq 0$), which is excluded when E_0 and F_0 do not commute, so no state attains both S_1 and S_2. (ii) If $\eta=0$, then $\Lambda=\kappa=\zeta$ and $\vec{r}=\hat{b}$ attains the bound. Otherwise, up to relabelling the outcomes of N, $\hat{b}=\hat{a}$, so $\Lambda=\eta+\zeta$, $\kappa=\max\{\eta,\zeta\}$, and $\vec{r}=\hat{a}$ gives $(p\oplus q)^{\downarrow}=\frac{1}{2}(1+\kappa,1+\Lambda-\kappa,1+\kappa-\Lambda,1-\kappa)$. (iii) Here $\kappa=0.95$ is reached only by the second outcome of N at $\vec{r}=-\hat{a}$, and $\Lambda=1.55$ only by $(s,t)=(1,-1)$ at $\vec{r}=\hat{a}$.

Proof of Proposition 4 and Corollary 1

(i) For $\mu=0$, $p=\eta p^\circ+(1-\eta)u$ with $u=(\frac{1}{2},\frac{1}{2})$, i.e. $p=Dp^\circ$ for the doubly stochastic matrix $D=\frac{1}{2}\left(\begin{smallmatrix}1+\eta & 1-\eta\\ 1-\eta & 1+\eta\end{smallmatrix}\right)$; likewise $q=D'q^\circ$. Hence $p\prec p^\circ$, $q\prec q^\circ$ and, since $D\oplus D'$ is doubly stochastic, $p\oplus q\prec p^\circ\oplus q^\circ$[28]. Schur-concavity gives the claim. (ii) By Theorems 1 and 4 both $\omega_\oplus$ and $\omega_\oplus(M,N)$ are lattice suprema. If $\omega_\oplus(M,N)\prec\omega_\oplus$, transitivity gives $p\oplus q\prec\omega_\oplus$ for all states; conversely, if $\omega_\oplus$ is an upper bound of all $p\oplus q$, then $\omega_\oplus(M,N)=\sup\prec\omega_\oplus$. Comparing ordered partial sums, $S_1=\frac{1}{2}(1+\kappa)\leq 1$ and $S_3=\frac{1}{2}(3+\kappa)\leq 2$ always hold, so the relation holds iff $S_2=1+\frac{1}{2}\Lambda\leq 1+\sqrt{c}$. When $\Lambda>2\sqrt{c}$ the state that attains S_2 (Theorem 4) gives $p_i+q_j=1+\frac{1}{2}\Lambda>1+\sqrt{c}$ for some i,j. (iii) For $\mu=\nu$, $\eta=\zeta$ and $\hat{a}\cdot\hat{b}\geq 0$, the terms of Λ are $2|\mu|+\eta|\hat{a}+\hat{b}|=2|\mu|+2\eta\sqrt{c}$ for $s=t$ and $\eta|\hat{a}-\hat{b}|\leq 2\eta\sqrt{c}$ for $s=-t$, so $\Lambda=2|\mu|+2\eta\sqrt{c}$, and $\Lambda\leq 2\sqrt{c}$ iff $|\mu|\leq\sqrt{c}(1-\eta)$. For click/no-click detection $|\mu|=1-\eta_d$, and the condition becomes $(1-\eta_d)(1-\sqrt{c})\leq 0$, which fails for $\eta_d<1$ and $c<1$.

Corollary 1: in a local-hidden-state model[12], $p(a,b|x,y)=\sum_\lambda p(\lambda)\,p(a|x,\lambda)\,\operatorname{tr}(\rho_\lambda K_b^y)$, where K_b^y are the effects of Bob's measurement $y\in\{M,N\}$. Conditioning on λ cannot increase entropy, and given λ Bob's outcome is independent of Alice's, so $H(B_y|A_x)\geq H(B_y|A_x\Lambda)=H(B_y|\Lambda)=\sum_\lambda p(\lambda)H(B_y)_{\rho_\lambda}$. Summing over the two pairs $(x,y)=(1,M),(2,N)$ gives $\sum_\lambda p(\lambda)[H(M)_{\rho_\lambda}+H(N)_{\rho_\lambda}]\geq\min_\rho[H(M)+H(N)]$, and Theorem 4 gives the last inequality in Eq. (12)[10,11]. For the example, $\hat{a}=\hat{z}$, $\hat{b}=\hat{x}$, $E_0=\eta_d|0\rangle\langle 0|$ and Bob in $|1\rangle$: $H(M)=0$ and $H(N)=h(\eta_d/2)$, independent of Alice's outcomes. For unbiased measurements, (i) shows that the ideal minimum is a valid right-hand side.

Proofs of Theorem 5 and Corollary 2

By the eigenvalue formula, S_k for the direct sum is the largest eigenvalue of a sum of k of the four projectors. Hence $S_1 = 1$ and $S_3 = 2$, since any three include $P_1 + P_2 = \mathbb{1}$ or $Q_1 + Q_2 = \mathbb{1}$, and $S_2 = \max_{i,j} \lambda_{\max}(P_i + Q_j) = 1 + \sqrt{c}$, since $\lambda_{\max}(P + Q) = 1 + \|PQ\|$ for any two projectors[66]. Moreover $c \geq \frac{1}{2}$: if $Q_1 \neq 0$ and ψ is a unit vector in its range, $\|P_1\psi\|^2 + \|P_2\psi\|^2 = 1$, so $\max_i \|P_iQ_1\|^2 \geq \frac{1}{2}$ (and if $Q_1 = 0$, then $c = 1$). The increments give $\omega_\oplus$, which is ordered. For the tensor product, $p_iq_j \leq (\frac{p_i+q_j}{2})^2 \leq w$, with equality at a top eigenvector of $P_i + Q_j$ with $\|P_iQ_j\|^2 = c$: by Jordan's lemma[47,66], $\mathbb{C}^d$ decomposes into mutually orthogonal subspaces of dimension one or two that are invariant under P and Q, and on a two-dimensional block with overlap c the bisector state of the induced qubit measurements gives $p_i = q_j = (1 + \sqrt{c})/2$. Hence $S_1 = w$, while a state in the range of P gives $S_2 = S_3 = S_4 = 1$. If $c < 1$, a state with $L_1(p \oplus q) = 1$ lies in the range of some P_i (or Q_j), so $q_j = \mathrm{tr}(\rho P_iQ_jP_i) \leq \|P_iQ_j\|^2 \leq c$ for all j and $L_2(p \oplus q) \leq 1 + c < 1 + \sqrt{c}$; likewise $L_2(p \otimes q) = 1$ requires two vanishing entries of $p \otimes q$, hence such a state, at which $L_1(p \otimes q) \leq c < w$. If $c = 1$, the ranges of some P_i and Q_j intersect and a common eigenvector attains both bounds. On each two-dimensional block $\{A, B\} = 2(\hat{m}_b \cdot \hat{n}_b)\mathbb{1}$ and $c_b = \frac{1}{2}(1 + |\hat{m}_b \cdot \hat{n}_b|)$, while $\{A, B\} = \pm 2$ on one-dimensional blocks, which gives $c = \frac{1}{2} + \frac{1}{4}\|\{A, B\}\|$[47].

For Corollary 2, S_1 and S_2 are the maximal partial sums by the eigenvalue formula (a pair within one measurement sums to 1, which never exceeds the cross terms, since $S_2 \geq S_1 + \frac{1}{2}$ as shown below), and $S_3 = 1 + S_1$ since any three effects include $E_1 + E_2 = \mathbb{1}$ or $F_1 + F_2 = \mathbb{1}$. The increments are non-increasing iff $S_1 + \frac{1}{2} \leq S_2 \leq 2S_1$. The upper inequality holds because $\lambda_{\max}$ is subadditive and $S_1 \geq \frac{1}{2}$; for the lower one, a top eigenvector of the effect that gives S_1 is a state at which the other measurement has an outcome of probability at least $\frac{1}{2}$. Lemma 1 completes the proof.

Numerical methods

All analytical results and every number quoted in the text were checked numerically with the code described under Code availability. Closed forms were compared with lattice suprema computed via least concave majorants over 1.8×10^5 grid states (40 random pairs of qubit bases, 60 random pairs of qubit POVMs; deviation $< 10^{-4}$) and with the eigenvalue characterization of S_k (200 random qubit POVM pairs; 150 random pairs of projectors of rank $d/2$ in $d = 2, 4, 6$; 300 random binary POVM pairs in $d = 2, \ldots, 6$; deviation $< 10^{-14}$). Pairs of projectors with a common eigenvector, which always occur unless both ranks equal $d/2$, have $c = 1$ and were checked separately. Exact minima of Rényi and Tsallis sums were obtained on a dense grid over the great circle in $\mathrm{span}\{\hat{m}, \hat{n}\}$, refined by Nelder–Mead; the reduction to this circle is justified by the radial argument above for unbiased measurements, and for biased ones by the concavity of the Shannon sum in (x, y), which is the only measure we minimize in that case. Near the eigenstates, where Rényi sums with $\alpha \leq \frac{1}{2}$ have a cusp, the outcome probabilities were written as $\cos^2(\varphi/2)$ and $\sin^2(\varphi/2)$ to avoid loss of precision. Proposition 4 was checked for 400 random biased POVM pairs (the criterion $\Lambda \leq 2\sqrt{c}$ against the largest Landau–Pollak sum over 8×10^3 states), for 100 random unbiased pairs (state-by-state majorization), and for the detector models of Fig. 6. The statements about light modes were checked in a Fock space of dimension 600.

Data availability

All data shown in the figures are generated by the code described below; no other data were used in this study. [TODO: Marcelo to confirm wording.]

Code availability

The code that reproduces every figure and checks every theorem and numerical statement (`majorization.py`, `verify_bounds.py`, `make_figures.py`; MIT licence) is available at [TODO: repository URL or Zenodo DOI; until the authors agree, "from the corresponding author upon reasonable request"].

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Author contributions

[TODO: e.g. R.Z. performed the analysis and numerics and wrote the manuscript; G.G. and M.F.C. conceived and supervised the project; all authors discussed the results and revised the manuscript.]

Competing interests

[TODO: The authors declare no competing interests.]

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